Upskilling Collaboration Projects

Names and brief conceptual plug for Dave Black's first set

Note that there is an alphabetized Abbreviation Guide on the page after this.

- Pocket Cocktail Party (Cocktail Party problem, Blind source separation of audio)
- Bilingual Boundary Lab (NLP models, performance without overfitting, Tolstoi)
- English Gorf (NLP looking at the distribution of Latin- and Germaic-based words)
- Envelope ZIP Reading (Digit recognition, computer vision, classic learning path)
- JabberVec-l2v (NLP, generative models for making gibberish words, vector encoding)
- Latent Forge (Synthetic data generation for ML recommendation-type predictions)
- Info Theory for Pareidolia/Apophenia (Academic Paper, functional information theory)
- CVMSS Paper Code Prep (Academic Paper, classification ontology, ML Computer Vision)
- Text Cleanroom (Encoding/decoding, Unicode, NLP additions like equivalency mapping)
- Multimodal Jupy Logger (Tooling: log important I/O cells, w/o complete Jupyter notebook)
- Mendeleev Spelling Bee (NLP, Depth First Search algorithm/data structure, letter-tiles)
- Start Pirate Now (R programming language through docs, expand to other coding langs.)
- Double-P Double-N (Stepping through Neural Networks, learn process & theory very well)
- Style Trajectories (Does ML attribute early/late works by a writer to one author?)
- Discrete Fourier Transforms and Sampling (Signal Processing, visualizing frequencies)
- Higgs Boson Visualized (Dataset visualization, analysis, and scientific discovery)
- RWKV + LoRA Learning Experiment (Nonstandard LLM fine-turning. Longer, complex)
- Job Application Word Frequency (NLP, data scraping, visualize/analyze results, improve)
- Cyrillic-through-Latin OCR (Create mobile-phone dataset creator for NLP translation)
- Multilingual Name Corruption Lab (NLP, LLM simulates likely errors → deeper search)
- Digital Sundial Learning (Science, representation, ML model learning design, digital sundial)
- MorphoGender (NLP, language comparison without learning language, language structure)
- Twain/Feynman Model Collapse (NLP, corpus, Generative writing, test degradation theory)
- Mechanics: Lagrange / Hamilton / T-V Simulations (Physics visualizations, coding graphics)
- Measuring Instruction Inconsistency (Heavier NLP, non-internal relevance to annotation)
- RLHF Incomplete Ontology (NLP, logic, analysis for things like rubrics, LLM Evaluation)
- Underground/Metro Systems Prediction Dataset (create novel dataset, do ML)
- Translated-Author Attribution Experiment (NLP, multilingual corpus, authorship attribution)

Abreviation Guide `CNN = Convolutional Neural Network ; HTR = Handwritten Text Recognition ; I/O = Input/Output ; ICA = Independent Component Analysis ; LLM = Large Language Model ; LoRA = Low-Rank Adaptation (for model fine-tuning) ; ML = Machine Learning ; MNIST = Modified National Institute of Standards and Technology dataset (of handwritten zip-codes) ; NLP = Natural Language Processing ; OCR = Optical Character Recognition ; PCA = Primary Component Analysis ; PPNN (Double P Double N) = Paper-and-Pencil Neural Networks ; RLHF = Reinforcement Learning from Human Feedback ; RWKV = Receptance Weighted Key Value ; SVD = Singular Value Decomposition ; w/o = without`

[Below is a summary of *`what the catalog is`* from Dave's ChatGPT.]

This is essentially a **menu of collaboration-sized technical projects**, ranging from things already producing research results to raw ideas that could become a weekend experiment. The common theme is that most are deliberately **small enough to understand and interrogate**, rather than "throw a giant model at a dataset."

The strongest through-lines are:

- **Interpretable ML / mathematical experiments.** *Latent Forge* studies recommender systems from means and low-rank approximations upward; *Pocket Cocktail Party* makes ICA/blind-source separation geometrically understandable; *JabberVec* examines the geometry of learned character embeddings to create gibberish words that sound like they could be English; *Digital Sundial Learning* scales a tiny learnable digital-sundial problem from 3 bits (possible light-path pixels) upward.
- **NLP and multilingual language structure.** *Bilingual Boundary Lab* studies Russian/French switching inside *War and Peace*; *English Gorf* looks for Germanic/French historical signals in English; *MorphoGender* asks how predictable grammatical gender is from noun form; proposed projects investigate translated-author attribution, multilingual name corruption, ambiguity, and a mobile dataset builder for Cyrillic OCR errors. This last one has the goal of creating a dataset to train a model that will take Cyrillic OCR results with Latin letters back to Cyrillic, e.g. `pe310Me 1A nopTqpo!7VIO` as mobile-phone output back to `резюме и портфолио` (meaning `resume and portfolio`.)
- **Computer vision + historical documents.** *CVMSS* is the substantial research lineage: detecting manuscript fragments reused in historical bindings, with later work involving high-resolution models, explainability, and experiment design. *Binding Unwinding* handles the decidedly unglamorous but important image-extraction/data-engineering side. *Info*

Theory for Pareidolia/Apophenia asks a particularly interesting foundational question related to CVMSS, which question looks at things independent of OCR/HTR results.

- **Small, conventional ML projects with room for serious analysis.** *Envelope ZIP Reading* offers MNIST neural-net/CNN work with emphasis on examining individual failures as well as solutions from different models. Not listed here are sources of concepts to rebuild independently, rather than code to recycle. If you want something more online-course-like, but with a twist to make it interesting as a GitHub portfolio object, ask Dave for a list of these.
- **Research infrastructure and engineering.** *CVMSS Paper Code Prep*, *Multimodal Jupy Logger*, and *Text Cleanroom* cover reproducibility, provenance, data cleaning, experiment infrastructure, packaging and practical software engineering. There is room for creating different UIs that might help with automation, though no such named project appears on the catalog list.
- **More speculative research seeds.** These include model collapse through repeated generated-data training, quantitatively measuring inconsistent instructions, incomplete ontologies in RLHF (Reinforcement Learning from Human Feedback [Editor's Note: This is the process used at such places as OpenAI and an organization that will be familiar to those of us working at Fort Point,] computational mechanics (simulating cool things that roll and swing and knock things over,) and inverse optics.
- **Dataset-building projects.** The worldwide *Underground Systems Prediction Dataset* (`underground` being the general name chosen for tunnel-based train systems known variously as metros, subways, tubes, undergrounds, etc.), *Multilingual Name Corruption Lab*, and OCR-pair collection (with *Cyrillic-through-Latin OCR*) are opportunities where constructing a genuinely interesting dataset is itself much of the intellectual work.

There is also a useful **maturity spectrum**. Some projects—especially CVMSS—represent a real research lineage. Latent Forge, Info Theory for Pareidolia/Apophenia, Bilingual Boundary Lab, Pocket Cocktail Party, JabberVec and several tooling projects are active experiments/prototypes. Others are deliberately **collaboration-ready seeds**, where someone could help define and run the first clean experiment.

So if I (this is Dave's ChatGPT talking) compressed the whole catalog to one sentence:

> It is a portfolio of small, inspectable ML/data/scientific-computing experiments—especially multilingual Natural Language Processing, interpretable mathematical Machine Learning, historical-document Computer Vision, and research tooling—designed so a collaborator can contribute something meaningful without first inheriting an enormous project.

And an important feature of the catalog is that **"upskilling" doesn't mean doing generic tutorial projects**. Even relatively elementary techniques—frequency vectors, PCA/SVD, MNIST CNNs, regression, matrix factorization—are attached to unusual questions or datasets, so someone can learn the technique while producing something that has a scientific or other practical reason to exist.